

Synthetic DCE from Non-Contrast Prostate MRI

Methods and Results Brief

Tier-1 static track — UCSF cohort

6 August 2026

Abstract

We predict a post-contrast DCE volume from three non-contrast sequences (T2w, DWI, ADC) on 3,667 UCSF prostate exams. A 3D conditional GAN reaches within-gland co-localization $r = 0.482$, against -0.043 for the strongest trivial baseline (copying the T2w channel), so the model predicts enhancement structure rather than passing anatomy through. The substantive result is a decomposition: **within-gland structure is predictable; per-patient enhancement amplitude is not**, and amplitude accounts for 51% of the signal. We show this is a property of the task rather than a tuning failure, and that the unidentifiable component is orthogonal to the PI-RADS curve-type endpoint, which is scale-invariant.

1 Task

Let $x = (x^{\text{T2w}}, x^{\text{DWI}}, x^{\text{ADC}}) \in \mathbb{R}^{3 \times D \times H \times W}$ be the stacked non-contrast input and $y \in [-1, 1]^{1 \times D \times H \times W}$ the target DCE phase. We learn $G_\theta : x \mapsto \hat{y}$ with $\hat{y} = \tanh(\cdot) \in [-1, 1]$. $\mathcal{M}_i \subset \Omega$ denotes the prostate mask of case i , $|\mathcal{M}_i| \approx 5\%$ of the crop.

2 Data and target selection

2.1 Cohort

Stage	Criterion	Remaining
Patients with DCE	—	5,439
Successfully staged	inputs resolvable, mask matches DCE grid	4,975
Enhancement QC (staging)	$\max_t \text{enh}(t) \geq 1.5$	4,246
DWI b-value filter	DWI resolves to $b \geq 600$	3,684
Enhancement QC recorded	—	3,667
Patient-level split		2805 / 312 / 550

Two cohort properties are findings rather than footnotes: $\approx 12\text{--}13\%$ of patients never enhance at any timepoint (failed or mistimed injection), and 13% of DWI resolves to $b=50$, which carries almost no diffusion weighting and therefore places a *different contrast* in the same input channel as the $b=1000$ the majority receive.

2.2 Phase selection by elapsed time

Enhancement relative to pre-contrast is

$$\text{enh}(t) = \frac{\mathbb{E}_{v \in \mathcal{M}} [S_v(t)]}{\mathbb{E}_{v \in \mathcal{M}} [S_v(0)]}, \quad (1)$$

measured as: flat at ≈ 1.0 to $t \approx 45$ s; wash-in 55–90 s ($1.12 \rightarrow 2.00$); then a **plateau** of 2.1–2.5 from ≈ 90 s past 300 s. There is no sharp peak, so $\arg \max_t \text{enh}(t)$ lands on plateau noise (≈ 280 s, into washout). Protocols are heterogeneous (26–70 phases, 1.7–13 s cadence), so a fixed *index* is a different physiological moment per patient. We select by *time*,

$$j^* = \arg \min_j |t_j - t_{\text{target}}|, \quad t_{\text{target}} = 120 \text{ s}, \quad (2)$$

which sits inside the plateau, past the timing-sensitive wash-in, at $\approx 92\%$ of maximum.

3 Preprocessing

Geometry. All modalities are resampled to a common physical grid (`--reference iso`) at spacing (0.47, 0.47, 3.0) mm, then center-cropped to $32 \times 192 \times 192$, i.e. a fixed field of view of $90 \times 90 \times 96$ mm for *every* patient. This matters: UCSF in-plane resolution spans 0.176–0.781 mm across 256–1024 matrices, so a fixed *pixel* crop of 256 covered anywhere from 40 mm to 320 mm of anatomy — an $8\times$ range of scale, with 172 cases cropped below the width of a prostate.

Intensity. Per-image (single center, so no cross-scanner alignment). For DCE, with body-tissue median m and robust spread s ,

$$y \mapsto \text{clip}\left(\frac{y - m}{k s}, -1, 1\right), \quad k = 1. \quad (3)$$

$k = 1$ was selected by measuring within-prostate target dispersion: $k=3 \Rightarrow 0.073$, $k=2 \Rightarrow 0.109$, $k=1 \Rightarrow \mathbf{0.218}$. Larger k flattens the target until ROI metrics become noise on a near-uniform field.

4 Objective

With ϕ a frozen 3D MedicalNet feature extractor,

$$\mathcal{L} = \lambda_1 \mathcal{L}_1^{\text{ROI}} + \lambda_s (1 - \text{SSIM}_{3D}(\hat{y}, y)) + \lambda_s (1 - \text{SSIM}_{\mathcal{M}}(\hat{y}, y)) + \lambda_p \|\phi(\hat{y}) - \phi(y)\|_2^2, \quad (4)$$

where the reconstruction term upweights the gland,

$$\mathcal{L}_1^{\text{ROI}} = \frac{\sum_{v \in \Omega} w_v |\hat{y}_v - y_v|}{\sum_{v \in \Omega} w_v}, \quad w_v = 1 + (\omega - 1) \mathbb{I}[v \in \mathcal{M}], \quad \omega = 10. \quad (5)$$

Without this the loss is dominated by background: the prostate is $\approx 5\%$ of the crop. For the latent diffusion models the objective is latent-space MSE, so ROI emphasis reaches them only through the first-stage reconstruction.

4.1 Flow matching

On the straight-line (rectified) path with $t = 0$ at the data and $t = 1$ at the source,

$$z_t = (1 - (1 - \sigma_{\min})t)z_0 + t\epsilon, \quad v^* = \epsilon - (1 - \sigma_{\min})z_0, \quad \epsilon \sim \mathcal{N}(0, I), \quad (6)$$

with $\sigma_{\min} = 0$ by default and $t \sim \mathcal{U}(0, 1)$ sampled per example, so the network learns the velocity field over the whole interpolation. The clean-latent identity

$$\hat{z}_0 = z_t - t v_\theta(z_t, t) \quad (7)$$

holds exactly for any $\sigma_{\min}$ and is what the optional image-space anchor decodes and supervises.

5 Metrics

All ROI statistics are computed **per case and then averaged**, never pooled over the batch. Pooling mixes between-patient brightness into every statistic: on data where per-patient brightness is predicted perfectly but within-gland structure is pure noise, pooling reports $r = +0.977$ where the truth is $+0.020$.

Localization (faithfulness). For case i with $\bar{y} = \mathbb{E}_{v \in \mathcal{M}_i}[y_v]$,

$$r_i = \frac{\sum_{v \in \mathcal{M}_i} (\hat{y}_v - \bar{\hat{y}})(y_v - \bar{y})}{\sqrt{\sum_{v \in \mathcal{M}_i} (\hat{y}_v - \bar{\hat{y}})^2} \sqrt{\sum_{v \in \mathcal{M}_i} (y_v - \bar{y})^2}}, \quad \text{roi_pearson} = \frac{1}{N} \sum_{i=1}^N r_i. \quad (8)$$

Texture and detail (realism).

$$\text{roi_var_ratio} = \frac{1}{N} \sum_i \min\left(\frac{\text{Var}_{v \in \mathcal{M}_i}[\hat{y}]}{\text{Var}_{v \in \mathcal{M}_i}[y] + \varepsilon}, 5\right), \quad (9)$$

$$\text{roi_grad_ratio} = \frac{1}{N} \sum_i \min\left(\frac{\mathbb{E}_{v \in \mathcal{M}_i} \|\nabla \hat{y}\|}{\mathbb{E}_{v \in \mathcal{M}_i} \|\nabla y\| + \varepsilon}, 5\right), \quad (10)$$

both with target value 1: below 1 is over-smoothed, above 1 is over-textured. The intensity-histogram distance uses order statistics of the masked voxels,

$$\text{roi_w1} = \frac{1}{2} \cdot \frac{1}{|\mathcal{M}_i|} \sum_k |\hat{y}_{(k)} - y_{(k)}|, \quad (11)$$

the $\frac{1}{2}$ mapping $[-1, 1] \rightarrow [0, 1]$. These combine into a label-free proxy

$$\text{realism} = \text{mean}\left\{\nu(\text{var_ratio}), \nu(\text{grad_ratio}), 1 - \min(\text{w1}, 1)\right\}, \quad \nu(x) = 1 - \min(|x - 1|, 1). \quad (12)$$

Fidelity. PSNR = $10 \log_{10}(L^2/\text{MSE})$ with $L = 2$ (the $[-1, 1]$ range). Global SSIM/PSNR are $\approx 99\%$ background and near-meaningless here; the *gap* between global and ROI values is the diagnostic.

6 The central result: an identifiability decomposition

6.1 Baselines

Three predictors that require no learning, with $\mu_i = \mathbb{E}_{v \in \mathcal{M}_i}[y_v]$ and $\bar{\mu} = \mathbb{E}_i[\mu_i]$:

$$\hat{y}_v^{\text{const}} = \bar{\mu}, \quad \hat{y}_v^{\text{level}} = \mu_i, \quad \hat{y}_v^{\text{t2w}} = x_v^{\text{T2w}}. \quad (13)$$

const knows nothing; **level** is an *oracle* that knows the correct per-case brightness but has no internal structure; **t2w** is the identity baseline.

Predictor	MAE _M	roi_pearson	Knows
const	0.2510	—	nothing
t2w copy	0.3372	−0.043	identity baseline
MODEL (3D GAN)	0.2355	+0.482	trained generator
level (oracle)	0.1851	—	correct brightness, no structure

6.2 Variance decomposition

Total ROI variance splits into a between-case (amplitude) and a within-case (structure) term:

$$\text{Var}_{i,v}[y] = \underbrace{\text{Var}_i[\mu_i]}_{\text{between-case, amplitude}} + \underbrace{\mathbb{E}_i[\sigma_i^2]}_{\text{within-case, structure}}, \quad \sigma_i^2 = \text{Var}_{v \in \mathcal{M}_i}[y_v]. \quad (14)$$

Measured on the test set: $\text{sd}_i[\mu_i] = 0.231$ and $\mathbb{E}_i[\sigma_i] = 0.226$, so

$$\frac{\text{Var}_i[\mu_i]}{\text{Var}_i[\mu_i] + \mathbb{E}_i[\sigma_i^2]} = \frac{0.231^2}{0.231^2 + 0.226^2} = \mathbf{0.51}. \quad (15)$$

6.3 Reading it

Structure is learned. The T2w copy scores ≈ 0 , so $r = 0.482$ is genuine localization and not laundered anatomy.

Amplitude is not. The model tracks per-patient brightness at only $\text{corr}_i(\mu_i, \hat{\mu}_i) = 0.334$ and compresses between-case spread $2.4\times$ ($0.231 \rightarrow 0.098$). Because amplitude is 51% of the variance, both models lose to the oracle-brightness baseline in MAE, and the flow model is worse than a *constant* predictor (-0.018) while still possessing real localization ($r = 0.369$).

Why this is the task, not the model. Per-case amplitude depends on gadolinium dose, injection rate, cardiac output, renal clearance and receiver gain — none of which is visible in T2w/DWI/ADC. We checked the DICOM headers for these variables: they were removed by de-identification (dose, rate, patient weight, field strength absent; manufacturer overwritten with "NA"). The information is not merely unused; it is not present.

Why it may not matter. PI-RADS curve types are scale-invariant: for any $\alpha > 0$,

$$\text{type}(\alpha C(t)) = \text{type}(C(t)), \quad (16)$$

since Type I (progressive), II (plateau) and III (washout) are defined by the *shape* of $C(t)$ after the peak. The unidentifiable component is orthogonal to the clinical endpoint.

7 Results

3D models, best checkpoint, held-out test ($n = 550$).

Metric	3D GAN	3D flow (MedVAE)
roi_pearson $\uparrow$	0.482	0.369
ssim_roi $\uparrow$	0.397	0.276
MAE $_{\mathcal{M}}$ $\downarrow$	0.236	0.269
roi_var_ratio $\rightarrow 1$	0.912	1.711
roi_grad_ratio $\rightarrow 1$	0.711	1.059
realism $\uparrow$	0.845	0.714
roi_pearson PZ / TZ	0.339 / 0.522	0.169 / 0.417

VAL $\approx$ TEST throughout. The GAN leads on faithfulness across four independent setups (2D and 3D, pixel and latent). The flow’s single advantage is texture (**grad_ratio** = 1.06 vs 0.71): it is less smooth, which reads as more realistic while localizing worse — the faithfulness/realism tradeoff in one row.

Peripheral zone. $r_{\text{PZ}} = 0.339$ against $r_{\text{TZ}} = 0.522$. PZ is where most clinically significant cancer arises and where DCE drives the PI-RADS upgrade rule, so this is the clinically weakest region and should be reported as such.

8 Reliability

8.1 Checkerboard artifact (fixed)

Transposed convolutions in all three decoders stamped a period-2 pattern into predictions. Detector: the lag-1 autocorrelation of the first difference along a row,

$$\rho = \text{corr}(d_k, d_{k+1}), \quad d_k = y_{k+1} - y_k, \quad (17)$$

which tends to -1 under a pure checkerboard and to -0.5 for white noise.

	ρ	HF power fraction
Real DCE target	+0.813	0.0003
GAN prediction (before)	−0.598	0.0015
GAN prediction (after)	+0.278	0.0003

The prediction alternated in sign at pixel scale *more strongly than white noise*. Removing it halved FID ($175.5 \rightarrow 88.9$), an independent confirmation.

8.2 Run-to-run variance

Two configuration-identical runs at the same seed differ. Training is not deterministic (cuDNN autotuning; non-deterministic **atomicAdd** in 3D conv backward), adversarial dynamics amplify the divergence, and best-checkpoint selection converts a bumpy validation curve into a noisy discrete choice of epoch.

Metric	Δ between identical runs
roi_pearson	0.017
roi_var_ratio	0.055
p75_corr	0.276

Rule. Differences below ≈ 0.02 in `roi_pearson` are unresolved at $n = 1$. *Safe:* GAN > flow (0.113, $\approx 7\times$ noise); model $\gg$ t2w ($\approx 30\times$); PZ < TZ (paired within run). *Not safe:* anything from `p75_corr` at $n = 1$; the iso-vs-no-iso comparison (0.020, inside noise).

Why p75_corr is the unstable one. `roi_pearson` averages $N = 550$ per-case values, so variance is suppressed by $\sqrt{N}$. `p75_corr` is a single cross-patient correlation, and it measures the *amplitude* dimension — precisely the component nothing in the loss constrains. **Reproducibility tracks identifiability:** the well-determined component reproduces, the ill-determined one drifts.

9 Tier-2: kinetics

The multi-timepoint target is the enhancement curve $C_t(\cdot)$ per voxel. Unlike generic video world models, the governing dynamics are *known* — the Tofts model,

$$\frac{dC_t}{dt} = K^{\text{trans}} C_p(t) - k_{ep} C_t(t), \quad k_{ep} = \frac{K^{\text{trans}}}{v_e}, \quad (18)$$

with C_p the arterial input function. A continuous-time latent world model (ODEWorld/PT-Flow) integrates a learned velocity field in *physical* time,

$$\frac{dz_t}{dt} = v_\theta(z_t, t; z_0, c) \implies z_T = z_0 + \int_0^T v_\theta(z_t, t; z_0, c) dt, \quad (19)$$

where z_0 encodes the initial (pre-contrast) state and c is a conditioning signal. Our contribution would be to *constrain* v_θ to the Tofts family with parameter maps predicted from bpMRI, rather than learn it free-form — a stronger inductive bias, guaranteed physically plausible curves, and a direct answer to the limitation that physics-informed ODE methods “require explicit prior knowledge of the governing equations,” which for DCE we have.

Goal state. Acquisition duration (measured, $n = 1367$): $p_5 = 222$ s, $p_{50} = 300$ s, $p_{95} = 380$ s. Coverage is 99.0% at 180 s, **91.9%** at 240 s, and only 49.7% at 300 s. We therefore anchor at $t = 240$ s: past plateau onset by ≈ 150 s, inside the window where Types II and III separate, while retaining 92% of the cohort.

10 Paper partitions and venue paths

The work separates into three claims with different evidence requirements, different readiness, and different venues. They are not sequential — ② is nearly complete on existing data while ① waits on labels.

	Claim	Still needs	Venue
①	Single-phase synthetic DCE is clinically usable; multi-vendor, PI-RADS endpoint	Hanxue’s labels; reader study; multi-seed repeats	Radiology / Lancet Dig. Health / Insights into Imaging
②	Standard synthesis metrics do not measure case-specific fidelity in ill-posed conditional generation	A within-model sweep; ≥ 3 seeds; a <i>second domain</i>	ICLR 2027 (tight) $\rightarrow$ ICML / NeurIPS
③	A physics-constrained velocity field beats a free-form one for contrast kinetics	Multi-phase staging (done); Tofts parameterization; interpolation ablation	NeurIPS world-models workshop $\rightarrow$ MICCAI 2027

① — **clinical.** The models exist and the pipeline is audited; the blocker is entirely the endpoint. The overall PI-RADS category is essentially absent from the report export (92/10,538 once a boilerplate footer URL is excluded), but the PI-RADS v2.1 *DCE component* is binary positive/negative and appears in $\approx 6,400$ reports, of which $\approx 2,374$ fall in our QC-passing cohort. That is the natural endpoint — it is the one component synthetic DCE is responsible for, and κ applies directly to a binary call. We are waiting on Hanxue’s labels rather than mining the free text ourselves.

② — **identifiability and evaluation.** The strongest near-term result, and it is *label-free*: the amplitude/structure decomposition (§5), the finding that both models lose to an oracle-brightness baseline while retaining real localization, and the observation that **reproducibility tracks identifiability**. The framing must avoid the perception–distortion tradeoff (Blau & Michaeli, CVPR 2018), which is established — the contribution is a *diagnostic for whether a conditional model used its conditioning at all*, together with evidence that the reported metrics cannot distinguish that case. A single medical domain will read as an application paper, so a second domain is the gating requirement, not a nicety.

③ — **kinetics.** The novelty is not porting a world model to a new organ (that is a MICCAI paper, and the source method’s authors would rightly be co-cited rather than superseded); it is that *we know the governing equation*. ODEWorld’s own related work identifies the requirement for “explicit prior knowledge of the true governing equations” as the limitation of physics-informed ODE methods — for DCE that prior knowledge is the Tofts model. The experiment is therefore free-form v_θ versus Tofts-constrained v_θ versus Tofts-plus-residual, under irregular temporal sampling.

Timing. Approximate and worth confirming: ICLR 2027 closes around late September 2026 (≈ 7 weeks out), NeurIPS workshops around September–October, MICCAI 2027 around February 2027 (≈ 6 months). Only ② is plausibly ICLR-ready in that window, and only if a second domain moves quickly; ③ fits the workshop now and a full venue in the MICCAI cycle. A rushed ICLR rejection would also consume the MICCAI window, so the go/no-go should be taken on the pre-contrast conditioning result rather than on ambition.

On the Tier-1 $\rightarrow$ Tier-2 gate. The project’s onboarding sets a hard gate: do not launch Tier 2 until PI-RADS $\kappa \geq 0.6$. That gate is currently *unmeasurable* rather than failed, since the overall

category is absent from the data. Two developments argue for revisiting it: the scarcity rationale behind it has evaporated (UCSF supplies $\approx 8\times$ the planned Tier-2 multi-phase pool at $\approx 10\times$ the temporal density), and curve-type classification is scale-invariant, so the one component we cannot predict is orthogonal to the Tier-2 endpoint. A defensible revision is to gate on *DCE-positivity* κ instead of overall PI-RADS κ .

11 Anticipated technical questions

Q1. Is `roi_pearson` = 0.482 inflated by *zonal* contrast rather than focal enhancement? A model that only learns “TZ is brighter than PZ” would score well without detecting anything. This is the sharpest objection and the data partly answers it. If the whole-gland correlation were driven by the PZ/TZ contrast, we would expect it to greatly exceed the within-zone values, since the between-zone term would be doing the work. It does not: $r_{\text{gland}} = 0.482$ sits *between* $r_{\text{PZ}} = 0.339$ and $r_{\text{TZ}} = 0.522$, and both within-zone correlations are solidly positive. So there is genuine within-zone structure, not merely zonal architecture. What this does *not* establish is lesion-level detection — reproducing normal zonal anatomy correctly still scores well. That requires lesion labels and a lesion-level readout, which we do not yet have.

Q2. You report a mean over 550 cases with no interval. What is the spread? Not yet reported, and it should be. `roi_pearson` is $\frac{1}{N} \sum_i r_i$; the per-case r_i distribution and a bootstrap CI over cases are computed from data we already have. Until then, the run-to-run figure (± 0.017) bounds *run* variability but says nothing about *case* variability, and the two should not be conflated.

Q3. Why Pearson? The signal–concentration relationship is nonlinear. Correct, and it is a real limitation. MR signal relates to gadolinium concentration through T_1 relaxation and the spoiled-gradient-echo signal equation, so a linear correlation measures the wrong thing if the nonlinearity is severe within the ROI dynamic range. Spearman ρ would be the rank-based alternative and is trivial to add. We report Pearson because it is what the synthesis literature uses; reporting both would be more defensible.

Q4. Does the 51% decomposition assume independence between the two terms? No. The law of total variance, $\text{Var}[Y] = \mathbb{E}_i[\text{Var}[Y \mid i]] + \text{Var}_i[\mathbb{E}[Y \mid i]]$, is an identity; no independence is required. One precision caveat: we substituted $(\mathbb{E}_i[\sigma_i])^2 = 0.226^2$ for $\mathbb{E}_i[\sigma_i^2]$. By Jensen $\mathbb{E}[\sigma^2] \geq (\mathbb{E}[\sigma])^2$, so the true within-case term is at least that large and 51% **is an upper bound on the amplitude share**. The qualitative conclusion — that amplitude is roughly half the signal — is unaffected.

Q5. Best checkpoints are selected on `ssim_roi`, which is smoothness-biased. Is “the GAN wins” partly a selection artifact? Partly, and this should be stated. `ssim_roi` rewards smooth predictions, and the GAN is the smoother model (`grad_ratio` = 0.71 vs 1.06). Selecting on *realism* or *balanced* would plausibly favour the flow. Two mitigations: the GAN also leads on `roi_pearson`, which is not smoothness-biased, and the margin (0.113) is $\approx 7\times$ the run-to-run noise. But a selection-metric ablation (same runs, re-selected under each criterion) is the clean answer and has not been done.

Q6. The LDM’s ROI weighting only enters through the first stage. Is the diffusion objective effectively unweighted? Yes, and it is a genuine asymmetry. The GAN’s reconstruction term is ROI-weighted at $\omega = 10$; the LDM trains on latent-space MSE with no clean mapping back to the mask, so ROI emphasis reaches it only via the frozen autoencoder’s reconstruction. Combined with a MedVAE first stage never trained on prostate DCE, the flow is handicapped in ways not isolated from its objective. “The GAN is better” is therefore a statement about these two configurations, not about GANs versus flows in general.

Q7. Were the 3D runs converged? 40 epochs versus 100 for 2D looks arbitrary. It was a wall-clock choice ($\approx 670\text{ s/epoch} \times 40 \approx 7.4\text{ h}$), not a convergence criterion, and convergence was not demonstrated. Against that: the GAN *degrades* late (discriminator wins, $d \rightarrow 0.03$, $\text{adv} \rightarrow 4.4$), and the selected best checkpoint clearly beats the final epoch (0.482 vs 0.419), so longer training is not obviously the missing ingredient. The `val_trajectory.jsonl` now logged per run will answer this directly.

Q8. Are the 12–13% “non-enhancers” biology, or registration failure? Not yet separated, and it is a real confound. Enhancement QC is computed as a mask-mean ratio, so a mis-registered mask would depress it exactly like a failed injection. We now capture a per-case registration quality (`reg` in `dce_times.json`, cohort median 0.459, $p_{90} = 1.694$) but have not yet correlated it against `enh_max`. If the non-enhancers concentrate in the poor-registration tail, the “failed injection” interpretation is wrong and the exclusion criterion needs rethinking.

Q9. FID on 550 grayscale MR volumes with an ImageNet Inception backbone — is that meaningful? Weakly. FID is biased at small sample sizes (KID is the low- n alternative), and Inception features are natural-image features. A RadImageNet or medical-SSL backbone would be the defensible choice. We use FID because it is the field standard and because it responded sharply and correctly to a known artifact (the checkerboard: $175 \rightarrow 89$), which is evidence it is sensitive to *something* real. It should not carry a headline claim.

Q10. realism_score is an unvalidated composite with equal weights. Agreed — it is a ranking heuristic for choosing between checkpoints without a reader session, not a validated metric, and no weighting study was done. It should not appear in a results table without that caveat; the components (`var_ratio`, `grad_ratio`, `w1`) should be reported individually.

Q11. var_ratio and grad_ratio are capped at 5. Doesn’t that bias the mean? Yes, by construction. The cap exists because a near-flat target ROI puts a small denominator in a ratio, and uncapped cases reached the thousands and destroyed the mean. The estimator is therefore biased toward 5 from above and should be reported with the median and the fraction of cases hitting the cap — neither of which we currently log.

Q12. Your target is normalized signal, not gadolinium concentration. How does that square with a Tofts-based Tier-2? It does not, cleanly. Tofts governs $C_t(t)$, and converting signal to concentration requires a baseline T_1 map (e.g. variable flip angle) that this cohort does not include. The workable fallback is relative enhancement, which makes the recovered parameters *semi-quantitative* — defensible if stated plainly, not if presented as K^{trans} in physical units. An arterial input function is a second unresolved requirement, and the 90 mm crop may not contain a usable artery.

Q13. Why not force determinism instead of measuring the noise? `torch.use_deterministic_algorithm` would raise on several 3D convolution kernels that have no deterministic implementation, and where it works it is substantially slower. More importantly it would suppress a real signal: the divergence is concentrated in the amplitude dimension precisely because nothing constrains it, so the instability is *informative*. The right response is ≥ 3 seeds with mean $\pm$ sd, not determinism.

Q14. A random 15% patient split of a single center — is the effective sample size really 550? Probably smaller. Protocols cluster (DCE PROSTATE, Siemens TWIST, GE DISCO), and the split is not stratified by protocol, so test cases are not fully independent draws. Stratifying the split by `series` — now captured in `stage_meta.json` — and reporting per-protocol metrics would be the fix, and would double as the multi-vendor generalization result the project promises.

Q15. How were the prostate masks produced, and were they checked? They are automatic segmentations delivered with the cohort (`pilot_log.json` records the strategy and resulting volume per case); we have not manually verified any of them. Every ROI metric in this document is conditioned on those masks being right, so mask error propagates directly into the headline numbers.

12 Caveats

2D and 3D ROI metrics are not comparable (the 2D dataset keeps only gland-containing slices, so the denominators differ). FID was measured on the last checkpoint, not the best. Every result predating the 2026-07 audit is superseded. No claim is made that additional data would close the amplitude gap: an earlier conclusion to that effect was measured on a configuration with known defects and has been retracted.